

Xihan Yao

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SUMMARY

GeoData/Remote Sensing researcher building reproducible, large-scale geospatial analytics. Experienced in land use/land cover modeling, spatial feature extraction, and multi-modal urban data (Street View, land cover, POI, mobility). Applies GeoAI and statistical modeling to study human–environment interactions, with strong Python-based pipelines and spatial diagnostics.

EDUCATION

- 2025 – present **The University of Texas at Austin** — PhD student, Geography and the Environment
Supervisor: Professor Yuhao Kang
Link: [GISense Lab](#)
- 2021 – 2023 **University of California, Berkeley** — MLA, Environmental Planning
Thesis: Towards more effective urban vegetation monitoring and management: Studying of urban tree characteristics and their relationship to the urban heat island effect with high-resolution remote sensing products, a case study in Portland, Oregon, USA.
Link: [Publication](#)
- 2017 – 2021 **Hong Kong University of Science and Technology (HKUST)** — BSc, Environmental Management and Technology
- 2019 **University of Washington, Seattle** — Exchange Student

EXPERIENCE

- The University of Texas at Austin — Research Assistant** Austin, TX, 2025 – present
- Leading a preregistered, national-scale research project modeling relationships between built environment characteristics and Big Five personality traits across U.S. urbanized areas.
 - Designing reproducible, Python-based workflows to process and harmonize TB-scale, multi-modal geospatial data, including Google Street View imagery, high-resolution land cover, POI data, and human mobility records.
 - Applying Earth foundation models (e.g., Google AlphaEarth) and statistical modeling to evaluate whether high-capacity embeddings provide additional explanatory power beyond traditional built-environment variables.
 - Conducting model diagnostics and spatial analyses (residual analysis, spatial autocorrelation tests) to identify geographic outliers, systematic failure modes, and limits of model generalizability.
 - Supervising four undergraduate research assistants and coordinating collaborative research across the lab and with external collaborators.
- EarthDefine, LLC — GIS Analyst** Redmond, WA, 2024 – 2025
- Designed and implemented deep learning–based raster classification workflows for city-scale, sub-meter land cover mapping and change detection, emphasizing reproducible batch processing and quality control.
 - Led a team of GIS specialists to deliver operational geospatial data products across the U.S., including community-level canopy coverage metrics and multi-year land-use change datasets; supported external stakeholders such as the Arbor Day Foundation and municipal governments.
- University of California, Berkeley** Berkeley, CA, 2022 – 2023
- Lecturer, GIS 2023
- Served as instructor of record for UC Berkeley’s largest GIS course (GEOG/LDARCH C188; 200 students),

- delivering lectures, designing assessments, and supervising a multi-TA instructional team.
- Redesigned and modernized the curriculum to incorporate contemporary GIS concepts and tools, working with ESRI to improve data workflows, reproducibility, and instructional data management.
- Graduate Student Instructor, Applied Remote Sensing & GIS** 2022 – 2023
 - Taught applied remote sensing labs using Google Earth Engine (JavaScript), covering object- and pixel-based image analysis, machine-learning land use classification, and multi-source data integration (Landsat, MODIS, NAIP, LiDAR).
 - Led hands-on GIS labs in ArcGIS Pro, updated legacy course materials, and produced instructional videos to support the transition from ArcGIS Desktop to ArcGIS Pro.

AWARDS

- **Energy AI Hackathon (6th Annual)**, UT Austin —Hildebrand Department of Petroleum and Geosystems Engineering (PGE).
Awarded **Fourth Place** in a competitive machine learning challenge (46 teams). Developed a "similarity-geospatial proximity gated model" achieving strong predictive accuracy and interpretability for subsurface production forecasting under sparse, heterogeneous conditions. 2026
- **Geoscience Hackathon**, UT Austin —Jackson School of Geosciences.
Awarded **Second Place**. Project: "Tracking Earth's Changes with AlphaEarth Foundations," applying foundation model embeddings for large-scale, flexible landscape system change detection. 2025
- **CyberTraining Summer School**, Texas A&M University —Department of Geography.
Awarded **First Place**. Project: "Assessing Instance-based Disaster Impact Through Multimodal Geospatial Data" 2025
- **IEEE GRSS GeoPitch Award**, IEEE Geoscience and Remote Sensing Society.
Awarded **Third Place**. Elevator Pitch Presentation: "Object-Based Urban Trees Characterization with Airborne LiDAR" 2023

SELECTED PUBLICATIONS

2. **Yao, X.**, Kim, M., Dronova, I., McBride, J. R., Kondolf, G. M., & Radke, J. D. (2025). Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification. *Landscape and Urban Planning*, 263, 105420. DOI: [10.1016/j.landurbplan.2025.105420](https://doi.org/10.1016/j.landurbplan.2025.105420)
1. Tong, J. C. K., **Yao, X. H.**, Lau, E. S. L., Cheung, W. K. S., Ho, K. K. S., Ng, V. Y. Y., & Lau, A. P. S. (2023). Light pollution impact assessment in Hong Kong: Multi-dimensional measurement and spatial numerical modelling on integrated light sources in the neighbourhood level. *Energy & Environment*, 35(5), 2497–2516. DOI: [10.1177/0958305X221146942](https://doi.org/10.1177/0958305X221146942)

SKILLS

Programming & Data Science	Python (TensorFlow, PyTorch, Scikit-learn, GeoPandas, Rasterio, PySAL), R, JavaScript (Google Earth Engine)
GIS & Remote Sensing	ESRI products (ArcGIS Pro, ArcGIS Online), QGIS, Google Earth Engine, Google AlphaEarth Foundations, LiDAR
Computing & Systems	Linux Shell, High-Performance Computing (HPC), GPU server management, batch processing, Git/GitHub